

Su Young Bae

About Myself.

Name	Su Young Bae
Email	sybae01@skku.edu
Github	https://github.com/BaeSuyoung
Linkedin	https://www.linkedin.com/in/suyoung99/
Scholar	https://scholar.google.com/citations?user=SoyBsicAAAAJ&hl=ko
Homepage	https://bsy99615.github.io/suyoungbae.github.io/
Interesting Topics	Trustworthy AI : Bias & Hallucination Mitigation, AI Ethics/Morality, Evaluation Robust and Efficient Learning : Robust Learning, Knowledge Retrieval, Reinforcement Learning, Efficient Tool Chaining & Memory Management Agentic AI : Multi-Agent System, Long Horizon Coding Agent, Text-to-SQL

Education

03/2024 – Present	Ph.D., Department of Artificial Intelligence Sungkyunkwan University, Suwon (GPA: 3.93 /4.5)
03/2022 – 02/2024	M.S., Department of Artificial Intelligence Sungkyunkwan University, Suwon (GPA: 4.13/4.5)
03/2018 – 02/2022	B.S., Department of Software Sungkyunkwan University, Suwon (GPA: 3.56 /4.5)

International Papers

05/2026	“Remember Your Trace: Memory-Guided Long-Horizon Agentic Framework for Consistent and Hierarchical Repository-Level Code Documentation” Suyoung Bae , Jaehoon Lee, Changkyu Choi, YunSeok Choi, Jee-Hyong Lee : <i>Preprint</i>
05/2026	PubTables-QA: A Multi-Page Table Visual Question Answering for Cross-Page Table Reasoning Jiin Han, Suyoung Bae , Yerim Choi, Sangyun Lee, Kyutae Kim, RyunHo Kim, Jee-Hyong Lee, Changkyu Choi, YunSeok Choi : <i>under review</i>
07/2026	“ReFEree: Reference-Free and Fine-Grained Method for Evaluating Factual Consistency in Real-World Code Summarization” Suyoung Bae , CheolWon Na, Jaehoon Lee, Yumin Lee, YunSeok Choi, Jee-Hyong Lee : The 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026 Main)
07/2026	“EXPO-SQL: Execution-based Clause-level Policy Optimization for Text-to-SQL” Jaehoon Lee, CheolWon Na, Suyoung Bae , Jin-Seop Lee, Jihyung Lee, YunSeok Choi, Jee-Hyong Lee : The 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026 Findings)
07/2026	“UCGRec: User-Centric Graph Learning for LLM-based Sequential Recommendation” CheolWon Na, HanBeul Kim, Suyoung Bae , YunSeok Choi, Jee-Hyong Lee : The 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026 Findings)

- 04/2026 “Adversarial Robustness in LLM-Based Multi-Agent Trading Systems: A Systematic Vulnerability Analysis”
CheolWon Na, **Suyoung Bae**, Jaehoon Lee, Jee-Hyong Lee
: **ICLR 2026 Workshop Advances in Financial AI**
- 04/2025 “DeCAP: Context-Adaptive Prompt Generation for Debiasing Zero-shot Question Answering in Large Language Models”
Suyoung Bae, YunSeok Choi, Jee-Hyong Lee
: Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (**NAACL 2025**)
- 04/2025 “SALAD: Improving Robustness and Generalization through Contrastive Learning with Structure-Aware and LLM-Driven Augmented Data”
Suyoung Bae*, Hyojun Kim*, YunSeok Choi, Jee-Hyong Lee (* : equal contribution)
: Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (**NAACL 2025**)
- 01/2025 “CharMoral: A Character Morality Dataset for Morally Dynamic Character Analysis in Long-Form Narratives”
Suyoung Bae, Gunhee Cho , Boyang Li, Yun-Gyung Cheong
: Proceedings of the 31st International Conference on Computational Linguistics (**COLING 2025**)
- 02/2023 “Predicting the Morality of a Character Using Character-Centric Embeddings”,
Su-Young Bae, Eun-Chong Kim, Yun-Gyung Cheong
: International Conference on Big Data and Smart Computing (**BIGCOMP 2023**)
- 09/2022 “The CreativeSumm 2022 Shared Task: A Two-Stage Summarization Model using Scene Attributes.”
Eun-Chong Kim, Tae-Woo Yoo, Gun-Hee Cho, **Su-Young Bae**, Yun-Gyung Cheong,
: International Conference on Computational Linguistics (**COLING 2022**)

Domestic Papers

- 11/2026 “RFScore: Evaluating Factual Consistency of Project-Level Code Summary Using Reference-Free and Fine-Grained Manner”
Suyoung Bae, CheolWon Na, Jaehoon Lee, Yumin Lee, YunSeok Choi, Jee-Hyong Lee
: Korean Artificial Intelligence Association Conference (**CKAIA 2025**)
- 08/2025 “CPGRec: Collaborative and Personalized Graph Learning in Large Language Models for Sequential Recommendation”
Hanbeul Kim, Cheolwon Na, **Suyoung Bae**, Yunseok Choi and Jee-Hyong Lee
: Korean Artificial Intelligence Association Summer Conference (**CKAIA 2025**)
- 04/2025 “Meaning-Preserving Query Rewriting and Query Reranking for Retrieval Augmented Generation”
Insung Hwang, **Suyoung Bae**, Jaehoon Lee and Jee-Hyong Lee
: KIIS Spring Conference (**KIISE 2025**)
- 04/2024 “RICE: Retrieval-Augmented Generation with Iterative Correction and Enhancement”
Doyeon Kim, **Suyoung Bae**, Jaehoon Lee and Jee-Hyong Le
: KIIS Spring Conference (**KIISE 2024**)
- 06/2022 “Can Language Model Learn Ethics? Predicting Characters’ Morality by Learning Character Embedding”
Su-Young Bae, Eun-Chong Kim, Yun-Gyung Cheong

07/2022 : Korean Artificial Intelligence Association Fall Conference (**JKIA 2023**)
 “Improving Document Retrieval Model Multi-Domain Dialogue Dataset”
Su-Young Bae, Eun-Chong Kim, Yun-Gyung Cheong,
 : Korea Computer Congress (**KCC 2022**)

Projects

07/2025 – Present	International collaborative research with Norway (Norway-Korea) AUroRA: Advancing Universal Resource-Scarce Large Vision-Language Models Towards Robust and Reliable AI
07/2025 – Present	Basic Science Research Program through the National Research Foundation of Korea (NRF) : A Trustworthy Multi-Agent Systems for User-Centric Project-Level Code Documentation (* project leader)
07/2025 – Present	AI Star Fellowship Support Program (IITP) : Development of Collaborative Intelligent Agent Technology for Human–AI Partnership, Participating Researcher
07/2024 – Present	Development of Open-Ended Alignment Fluxional AI for Ever-changing Environment and Value (ITRC) : Developing open-ended alignment AI, Participating Researcher
04/2024 – Present	Developing Multimodal Generative AI Talent for Industrial Convergence (IITP with 42MARU) : Development of Adaptive Text Generation Technology Reflecting the User's Multi-faceted Persona, Participating Researcher
2022 2023 2024 2025	Creative Autonomous Research Project (S.E.L.P) : Various Topics on Story character embedding, Story character ethics, Long-Text Summarization, Multimodal bias mitigation, Hallucination detection in LLMs
03/2022 – 12/2022	Collaborative story-making tool development project (NCSOFT) : Character Analysis, story creation tool, and graph model development.
03/2021 – 07/2021	Undergraduate Research Project (URP) : GPT2 generated sentence qualification using the ROCStories dataset.
05/2019 – 12/2019	Co-operation Project (Rambda 256) : Develop Sungkyunkwan University blockchain coin mobile service (SKKoin).
05/2018 – 12/2018	Co-operation Project (SKT) : Marketing, Data Collection, Preprocessing, Analyzing

Internship

08/2026 – 10/2026	Oslo University in Norway Research Internship
06/2023 – 08/2023	Nanyang Technological University (NTU) Research Internship
03/2021 – 02/2022	Sungkyunkwan University (SKKU) Undergraduate Research Internship

Award & Scholars

02/2026	Sungkyunkwan University Selection for Incentives for outstanding activities (1200\$)
02/2026 – Present	Scholarship for outstanding AI graduate students (1500\$ per semester)
09/2025 – 09/2027	Ph.D Fellowship in the National Research Foundation of Korea (NRF) (18000\$ per year)

09/2025 – 02/2026	Scholarship for outstanding AI graduate students (3000\$ per semester)
05/2024 – Present	Scholarship for outstanding Ph.D. students (3600\$ per year)
02/2023	Sungkyunkwan University Selection for Incentives for outstanding activities
10/2022 – 02/2024	Sungkyunkwan University BK21 Education Research Scholarship (8600\$ per year)
2022 2023 2024	Sungkyunkwan University AI Selection of Excellent Team for Creative Autonomous Research (S.E.L.F.)

Assistant & Reviewer

2025	NeurIPS, AAAI, EMNLP, ACL Reviewer
12/2024 – 02/2025	Teaching Assistant (TA) for the Undergraduate Research Project (URP)
2022 2023	Teaching Assistant (TA) for the Computer Programming for Engineers course
09/2021 – 12/2021	Teaching Assistant (TA) for the Problem -Solving Algorithm course

Language study abroad

01/2020 – 01/2020	Software Silicon Valley IT Visiting Program
02/2020 – 06/2021	Summer Exchange Student (Czech Technical University)

English skills

02/2023	OPic	IH (Intermediate High)
09/2021	GRE	Verbal/Quantitative/Writing: 144/168/3.0